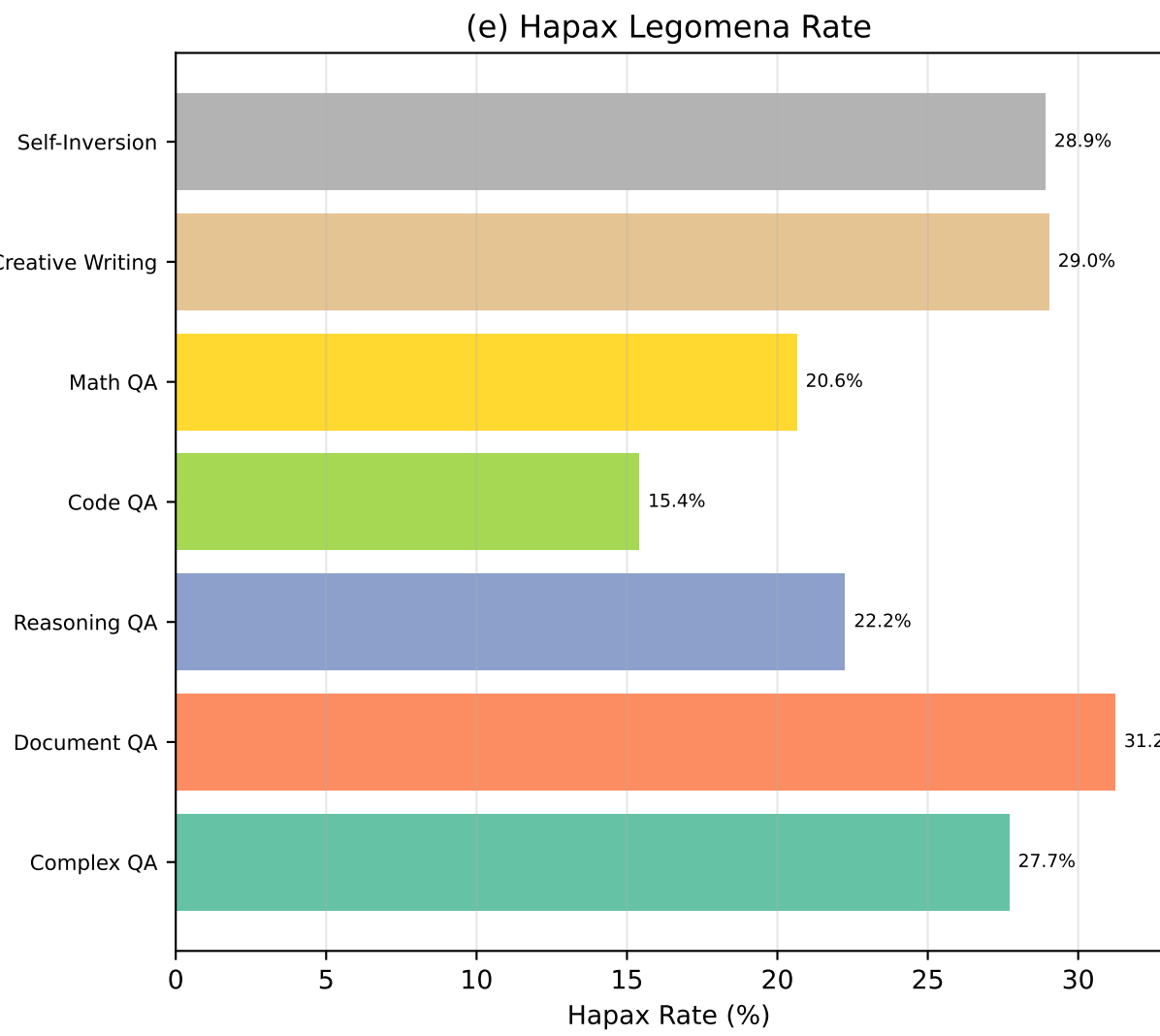
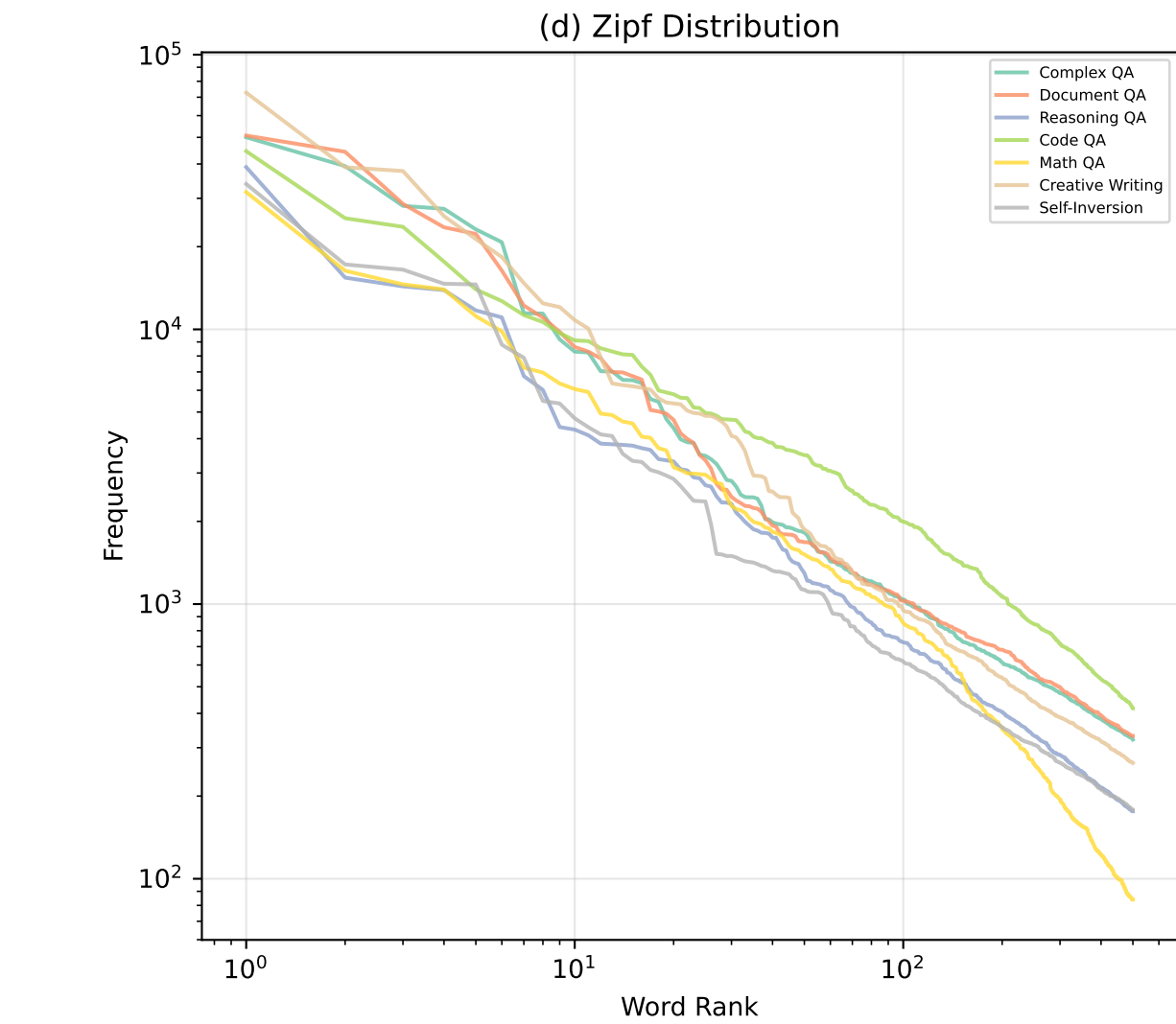
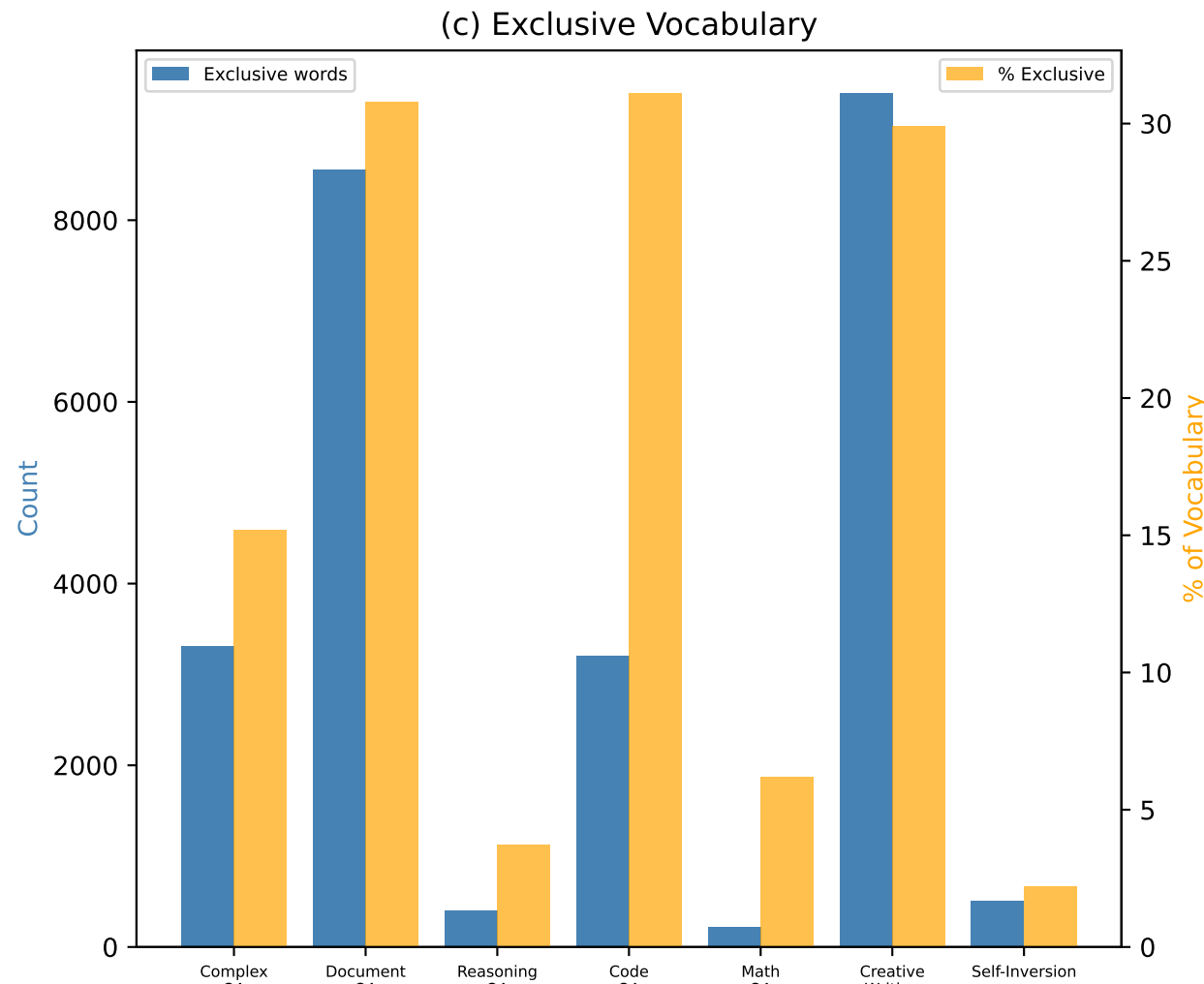
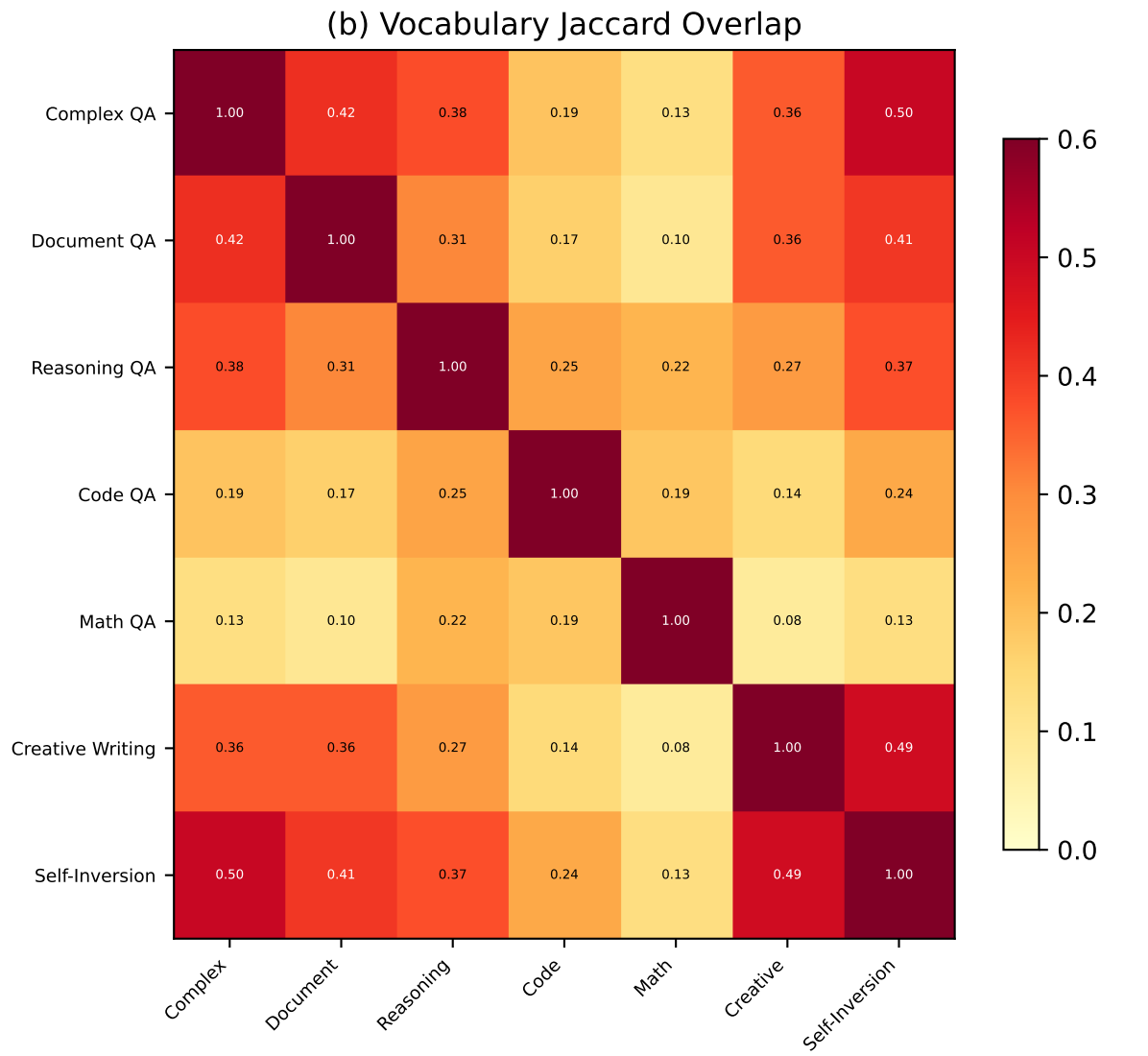
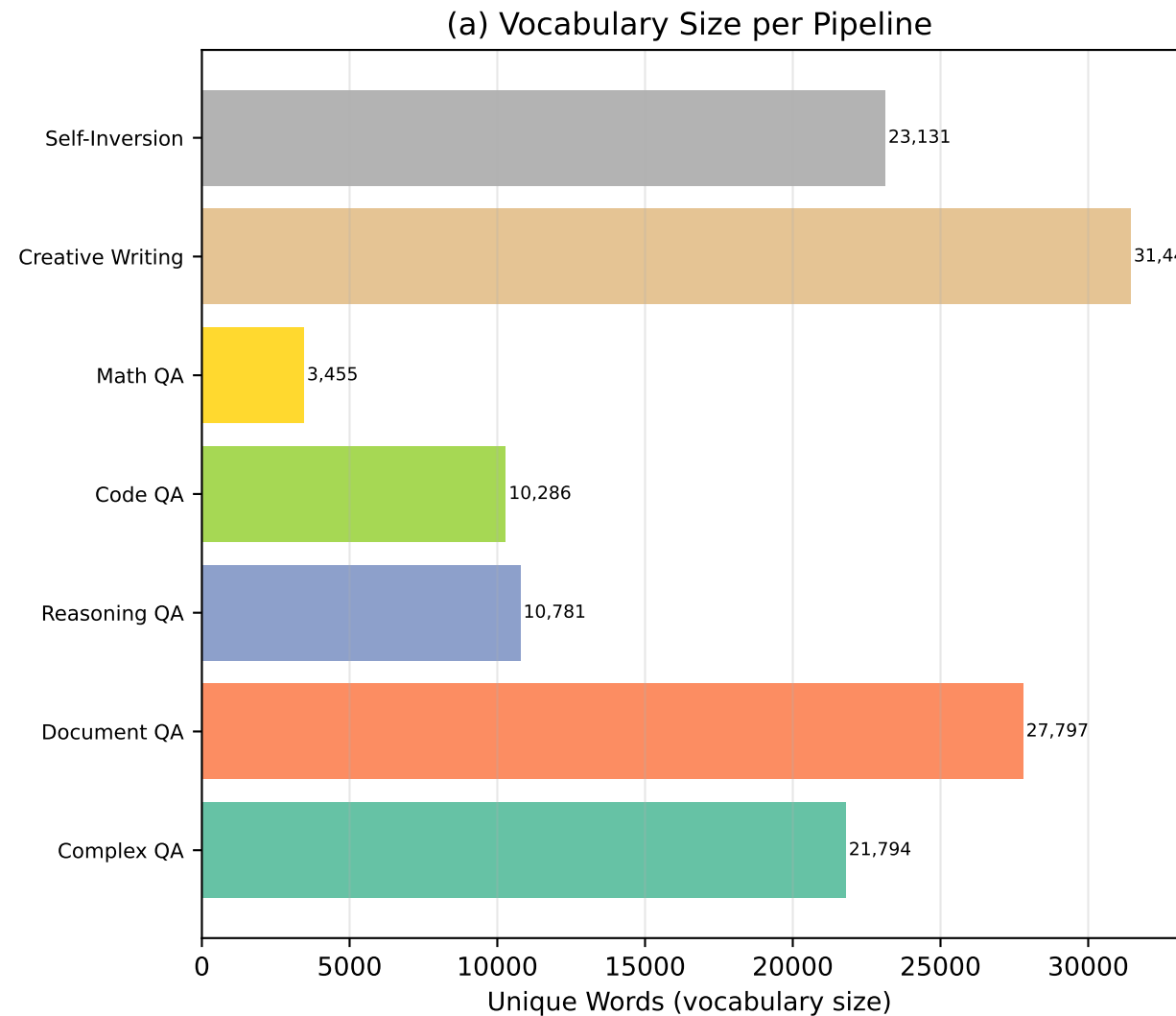


Fig 136: Token Distribution & Vocabulary Ecosystem Analysis



(f) Vocabulary Ecosystem Summary

| Metric                  | Value            |
|-------------------------|------------------|
| Total unique words      | 54,152           |
| Shared by ALL pipelines | 1,841 (3.4%)     |
| Avg vocabulary/pipeline | 18,383           |
| Avg exclusive/pipeline  | 3,656            |
| Avg hapax rate          | 25.0%            |
| Max vocab               | Creative Writing |
| Max exclusive           | Creative Writing |
| Max hapax               | Document QA      |